

INFO 5602
Information Visualization

“Choosing an Effective Visual”

We are NOT naturally good at **storytelling** and **presentation structuring**

Presentation = Noise + Signal

- *Clutter/distractions on slides*
- *Stakeholder - irrelevant information included*
- *Stakeholder -irrelevant “design” options*
- *Distracting colors, objects included in presentations*
- ***Adds cognitive load to stakeholders***

Find + Remove

- *Stakeholder-relevant information*
- *Stakeholder-relevant charts*
- *Stakeholder-relevant design options*
- *User-centric design for charts*
- ***Improves effective stakeholder communication***

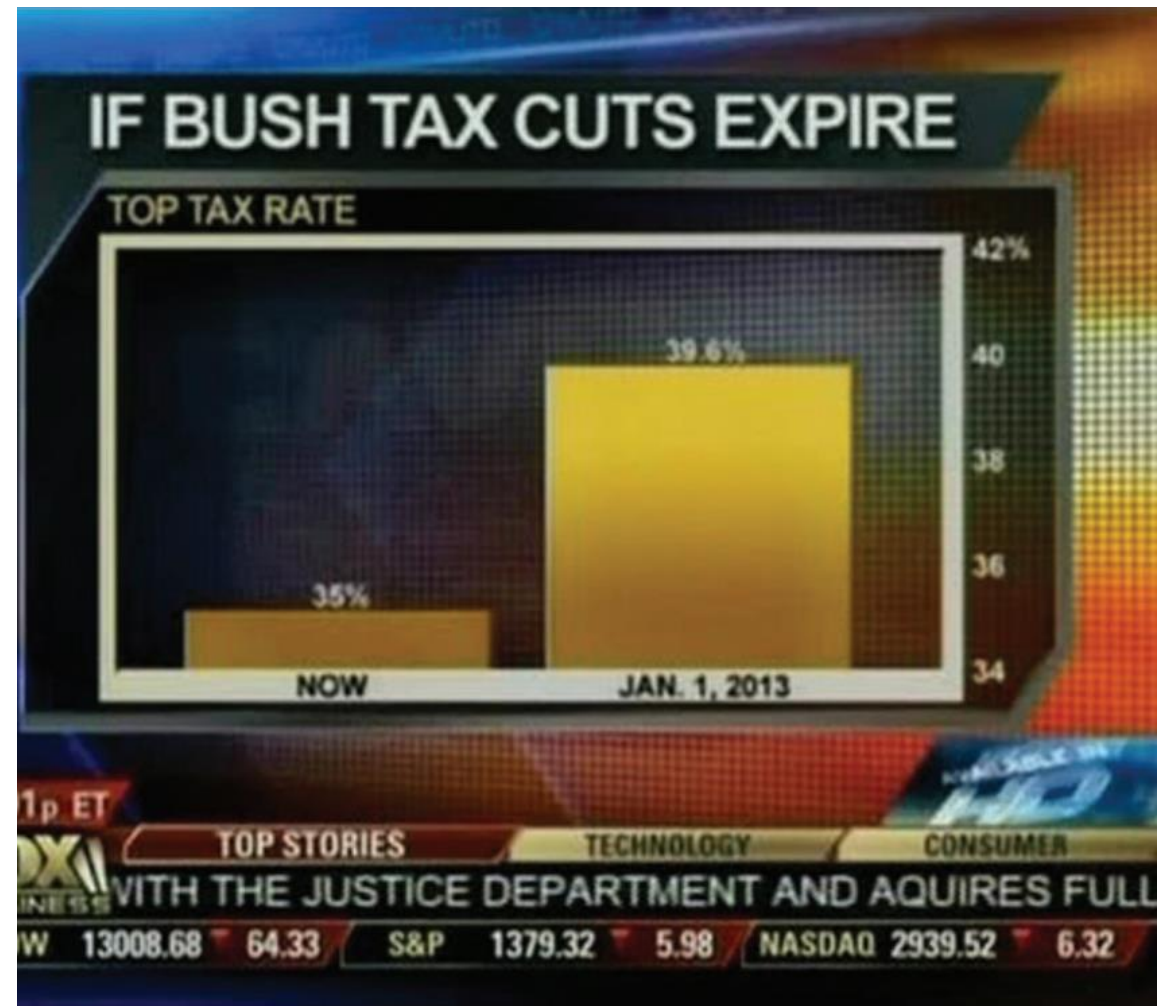
Find/Create + Boost

Agenda: the key charts you will use professionally

- Simple text
- Tables
- Heatmap
- Graphs
- Points
 - Scatterplot
 - Line graphs
 - Slope graphs
- Bar charts
- Area charts
- “Pie charts are evil”
- “Never use 3D”
- Is there a “most effective” visual?

Bar charts

- Bar charts should be used as much as possible – because they are common and mean **less of a learning curve** for your audience (lower cognitive load)
- Our **eyes immediately compare the relative end points of the bars**; for this reason, to the extent possible, bar charts should always have a zero baseline (where the x-axis crosses the y-axis at zero), otherwise you get a false visual comparison

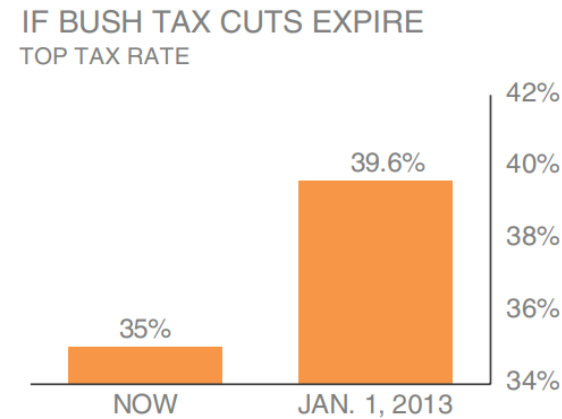


2.12 Fox News bar chart

Bar charts

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- Our eyes immediately compare the **relative end points of the bars**; for this reason, to the extent possible, **bar charts should always have a zero baseline (where the x-axis crosses the y-axis at zero)**, otherwise you get a false visual comparison
- Line graphs **may not always** have a zero baseline

Non-zero baseline: as originally graphed



Zero baseline: as it should be graphed

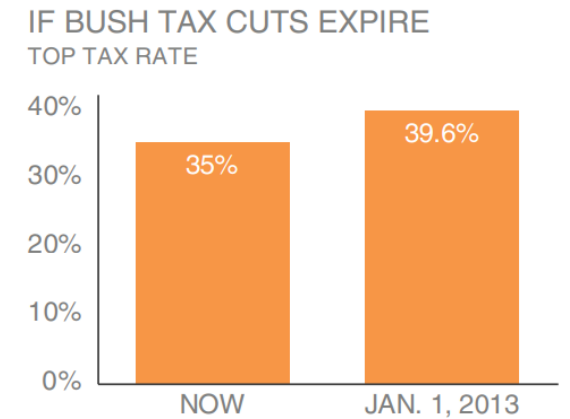
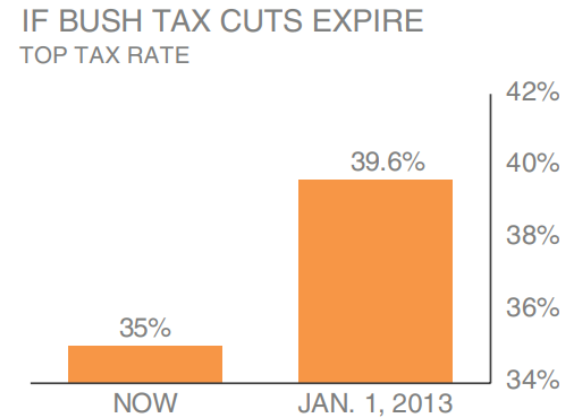


FIGURE 2.13 Bar charts must have a zero baseline

Bar charts

- Other design changes to reduce cognitive load:
 - The y-axis placed on the right. We read information from left to right (Z-manner)
 - Data labels originally outside the bars were pulled inside the bars
 - For your audience to benefit from focusing on big picture trends, think about preserving the axis but de-emphasizing it by making it grey

Non-zero baseline: as originally graphed



Zero baseline: as it should be graphed

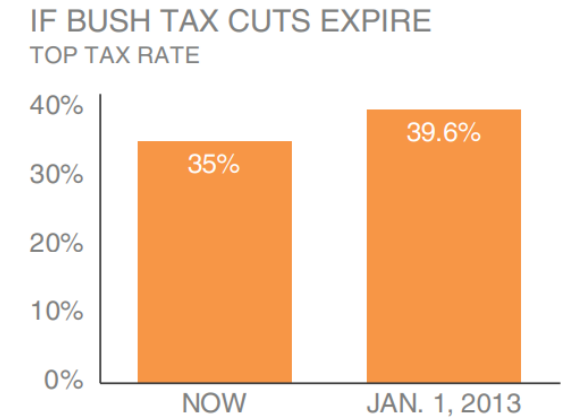
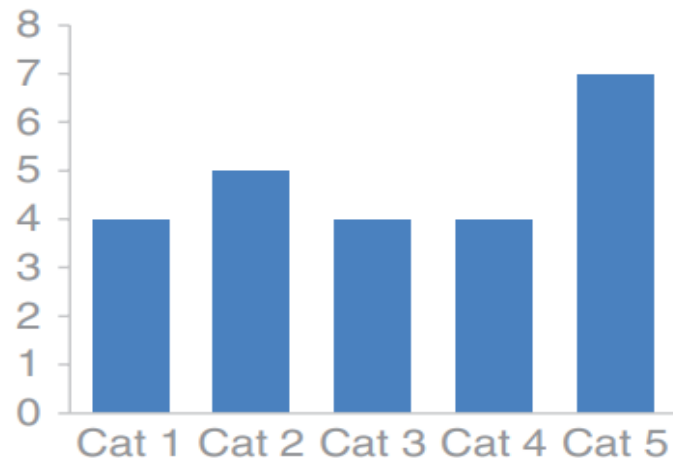


FIGURE 2.13 Bar charts must have a zero baseline

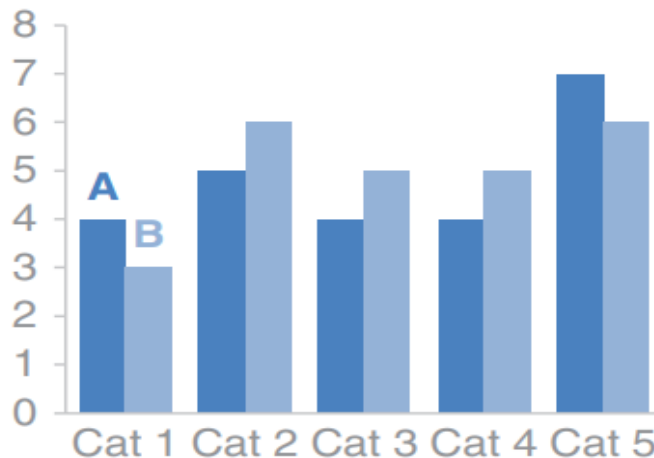
Bar charts: vertical bar charts

- Like line charts, vertical bar charts can be single series, two series or multiple series. As you add more series of data, **it becomes more difficult (a.k.a. cognitive load) to focus on one series at a time and obtain insight** – so use multiple series bar charts with caution
- Use the relative order of categorization cautiously – **emphasize relatively important categories/series** and de-emphasize unimportant information

Single series



Two series



Multiple series

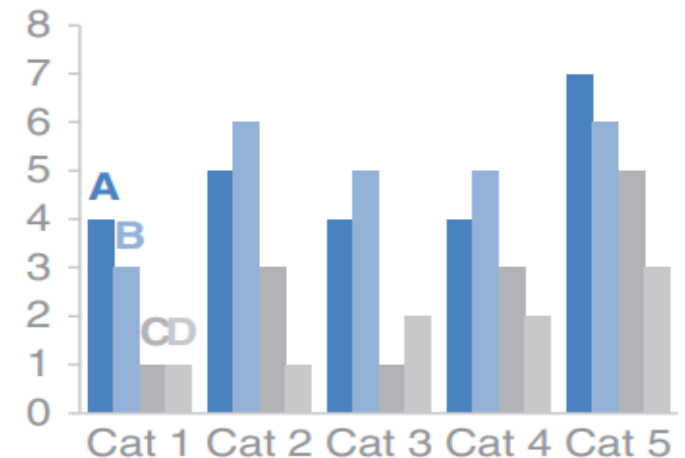
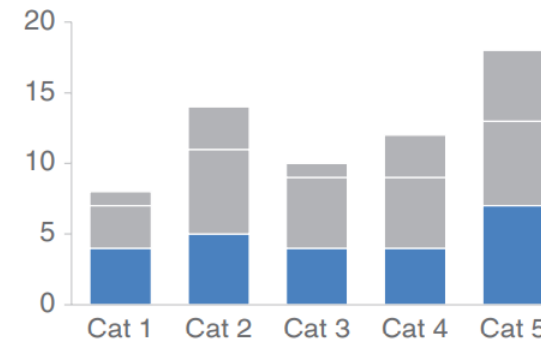


FIGURE 2.15 Bar charts

Bar charts: stacked vertical bar charts

- Stacked vertical bar charts are used to compare totals across categories and also see subcomponent pieces within a given category
- Two variants: **absolute numbers stacked vertical bar chart** and **100% stacked vertical bar charts**
- It can be difficult to compare sub-component across the various categories in stacked vertical bar charts once you get beyond the bottom sub-component as shown here (**absolute numbers stacked vertical bar chart**):

Comparing **these** is easy



Comparing **these** is hard

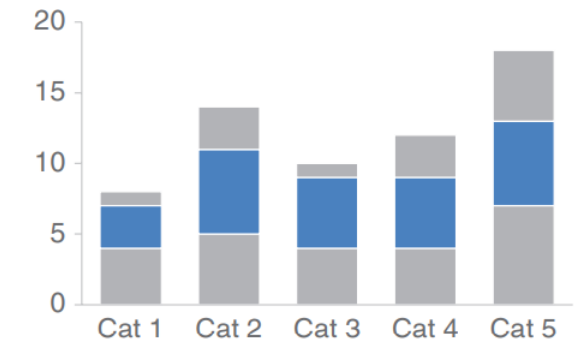


FIGURE 2.16 Comparing series with stacked bar charts

Bar charts: waterfall chart

- Waterfall charts can be used to pull apart the pieces of a stacked bar chart to focus on one at a time, or to show a starting point, increases and decreases, and the resulting ending point
- Waterfall charts help your stakeholders/ audience see a progression of steps or activities (and their impact on what you are tracking) and **aid your storytelling**
- In the example across, we are **tracking headcount**

2014 Headcount math

Though more employees transferred out of the team than transferred in, aggressive hiring means overall headcount (HC) increased 16% over the course of the year.

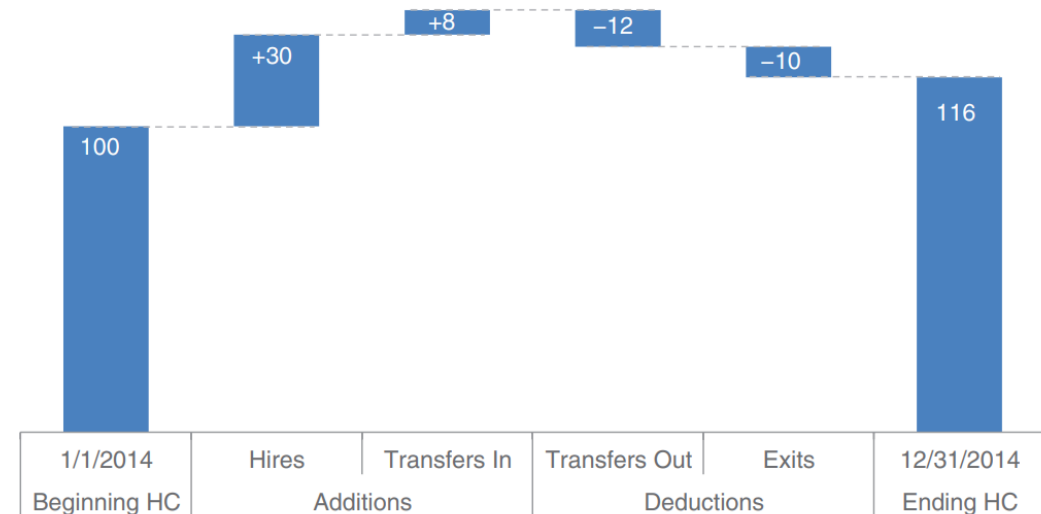
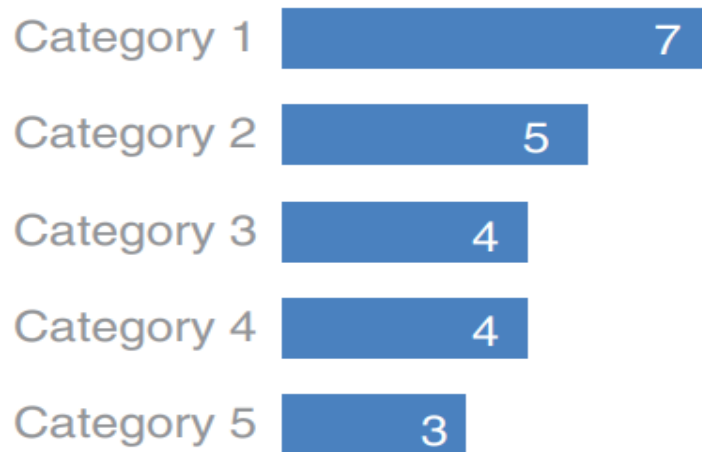


FIGURE 2.17 Waterfall chart

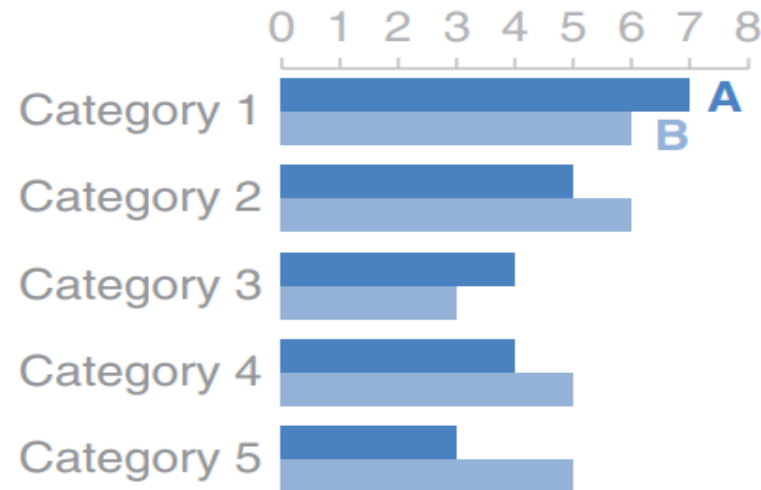
Bar charts: horizontal bar charts

- Relative to vertical bar charts, horizontal bar charts are easier to read
- Horizontal bar charts are **especially useful if your category names are long** – as the text is written from left to right, as most audiences read. This makes your graph easy to consume (a.k.a lower cognitive load) by your stakeholders

Single series



Two series



Multiple series

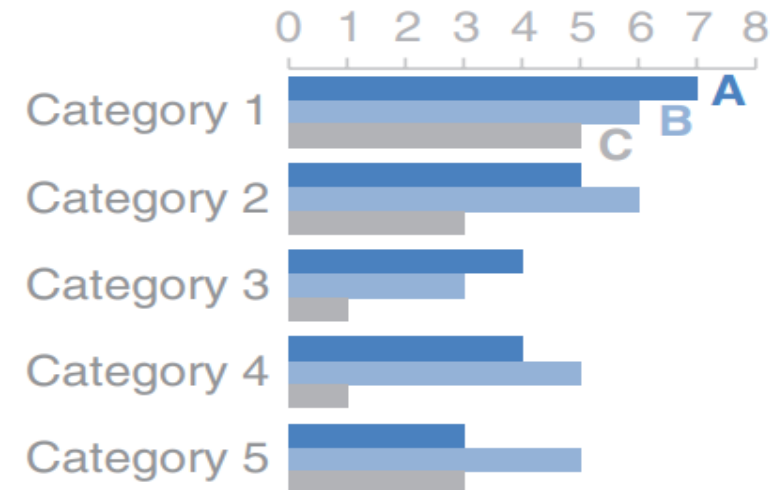
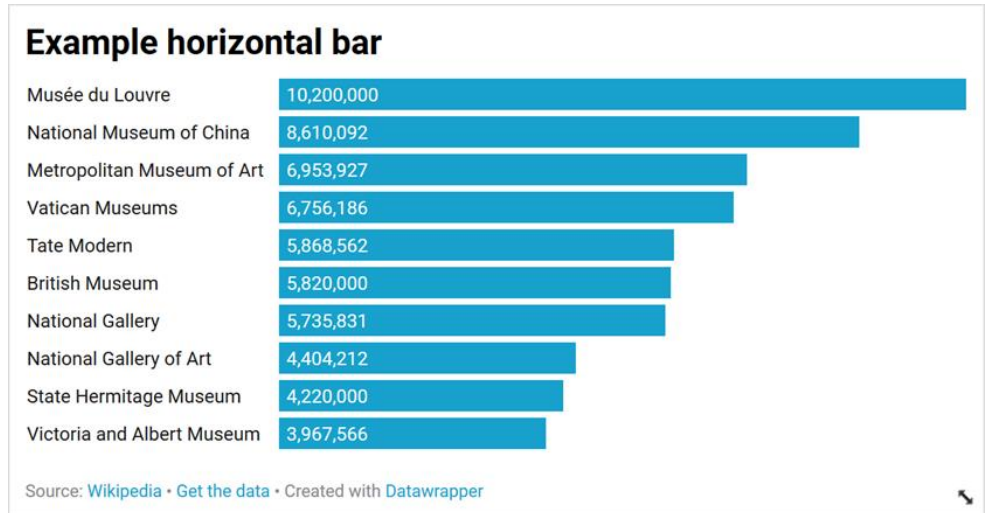
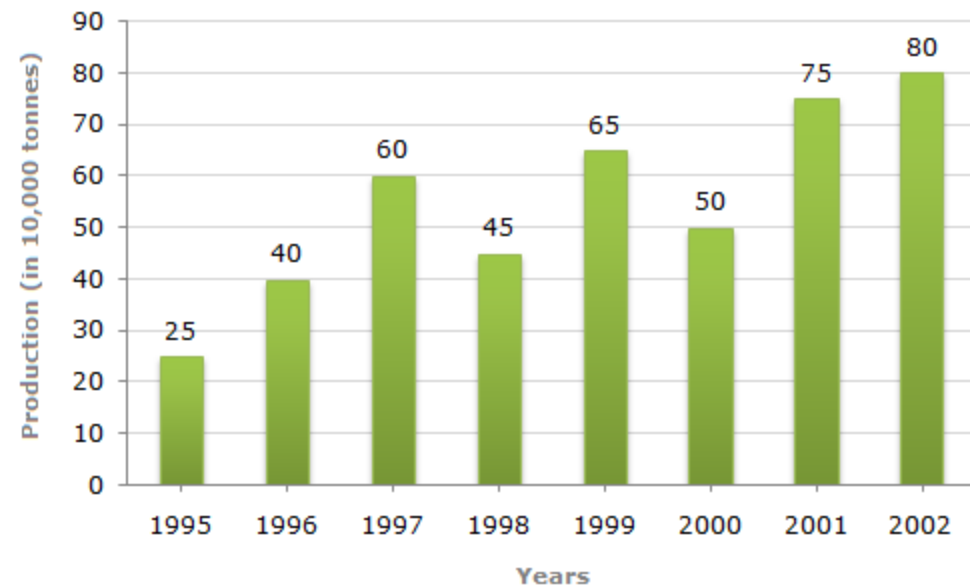


FIGURE 2.18 Horizontal bar charts

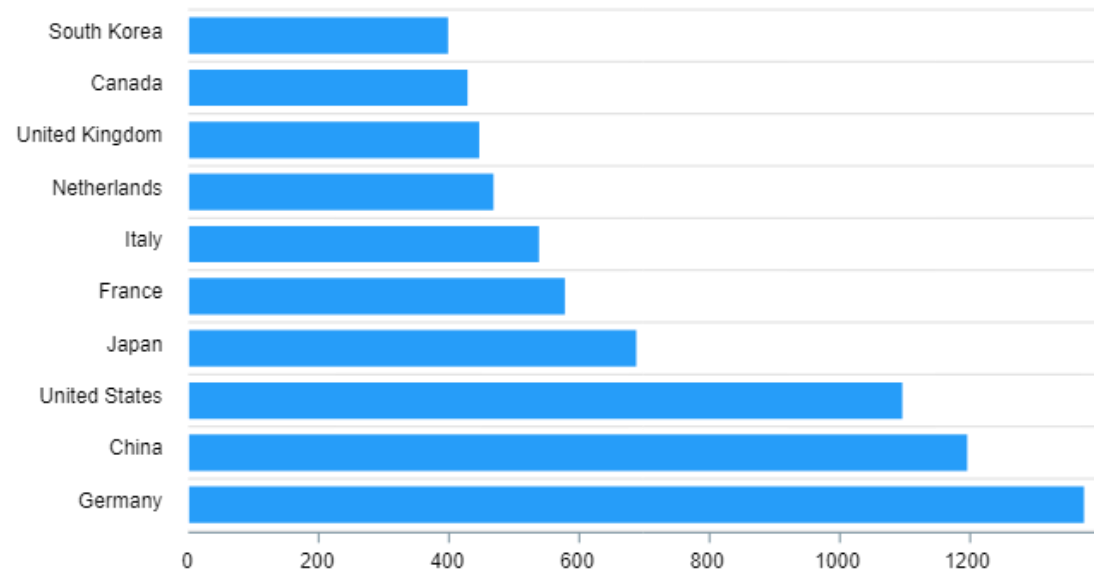
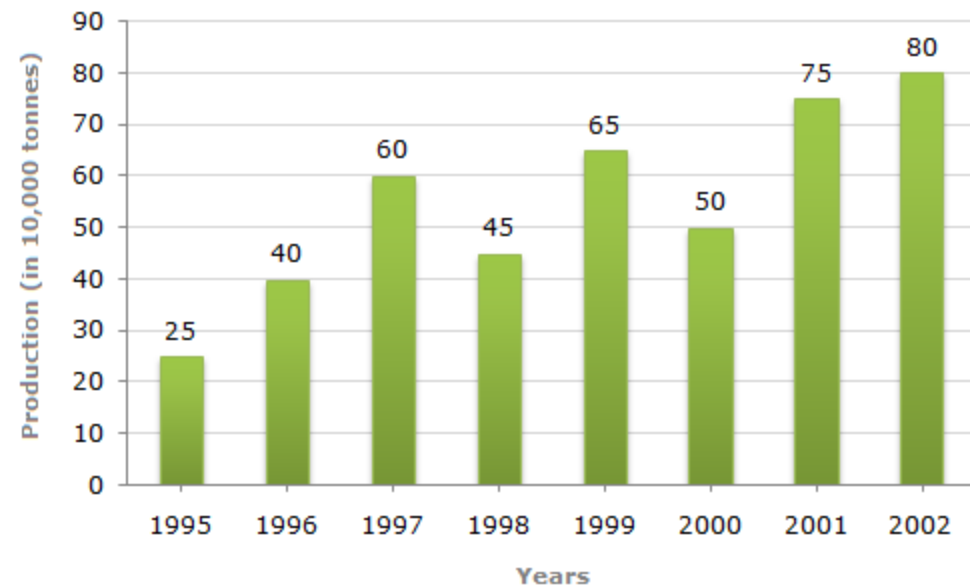
Logical ordering of bar charts

- When the category axis is **time**, arrange bars **chronologically**
- When the category axis is **not time**, arrange in **ascending** or **descending** order of magnitude of the bars and de-emphasize irrelevant information



Logical ordering of bar charts

- When the category axis is **time**, arrange bars **chronologically**
- When the category axis is **not time**, arrange in **ascending** or **descending** order of magnitude of the bars and de-emphasize irrelevant information



Bar charts: stacked horizontal bar chart

- Similar to stacked vertical bar chart, stacked horizontal bar charts are used to show totals across different categories but also give a sense of the subcomponent pieces
- Stacked horizontal bar charts can be structured to show either **absolute values** or **sum to 100%** (shown across):

Survey results

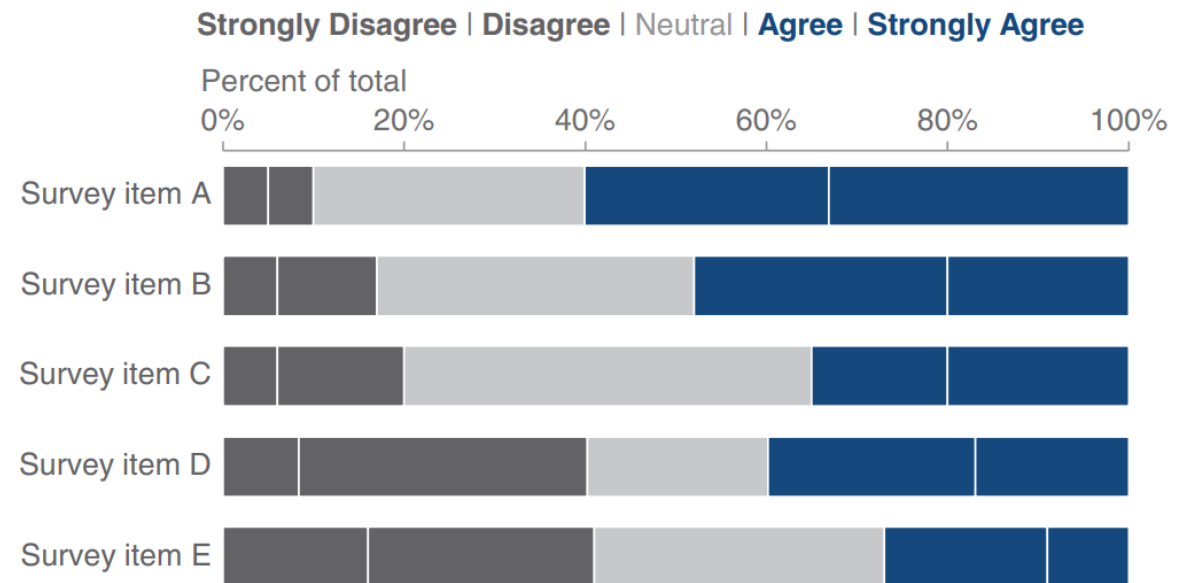
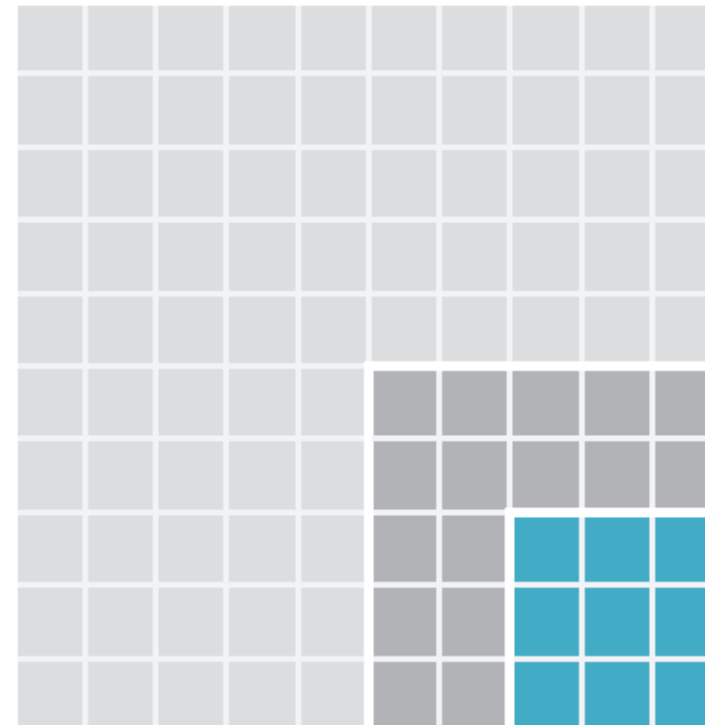


FIGURE 2.19 100% stacked horizontal bar chart

Area charts

- Human eyes do a poor job of attributing quantitative value to two-dimensional spaces, which can render area graphs harder to read than other charts
- Area charts are useful if your audience is receptive to visualizing numbers of vastly different magnitudes
- **Annotations** can be used to **reduce stakeholder cognitive load** when it comes to reading area charts (as shown here)

Interview breakdown



Out of every **100**
phone screens.

we bring **25**
candidates on
for interviews...

and
extend 9 offers.

FIGURE 2.20 Square area graph

“Pie charts are evil” - Knafllic (2017)

- Thought experiment: which supplier has the largest market share?

Supplier Market Share

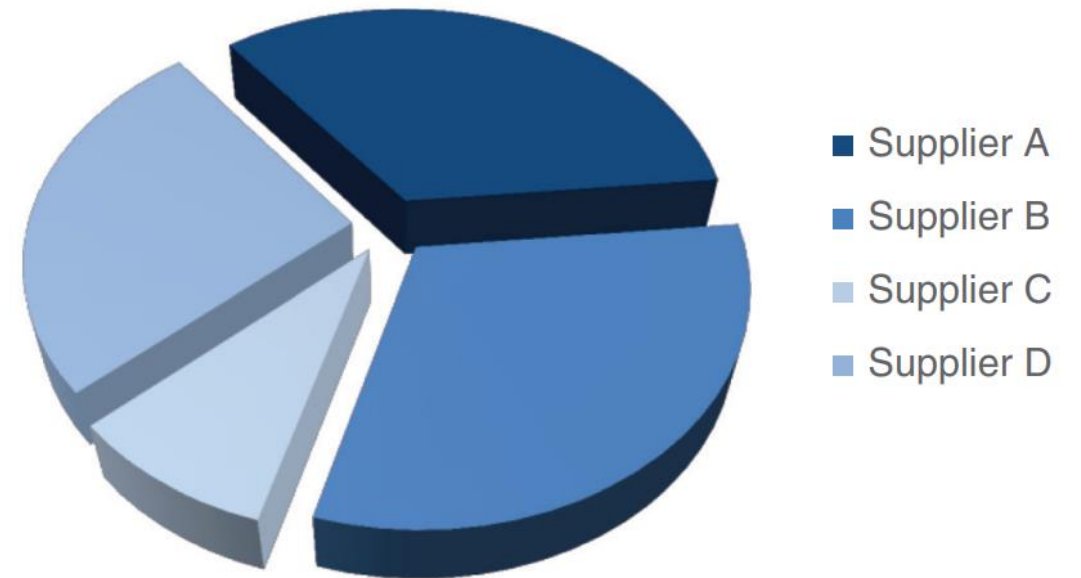


FIGURE 2.21 Pie chart

“Pie charts are evil”

- Knafllic (2017)

- Thought experiment: which supplier below has the largest market share?
- A pie chart is another example of a two-dimensional space. **Without labels or annotations**, human eyes do a poor job of ascribing quantitative value to two dimensional space
- Pie charts are hard (a.k.a cognitive load) for people to read for the following reasons:
 - When pie chart **segments are close in size**, it is difficult to tell which is bigger.
 - When segments are not close the best you can do is determine that one is bigger than the other – **but you cannot judge by how much**

Supplier Market Share

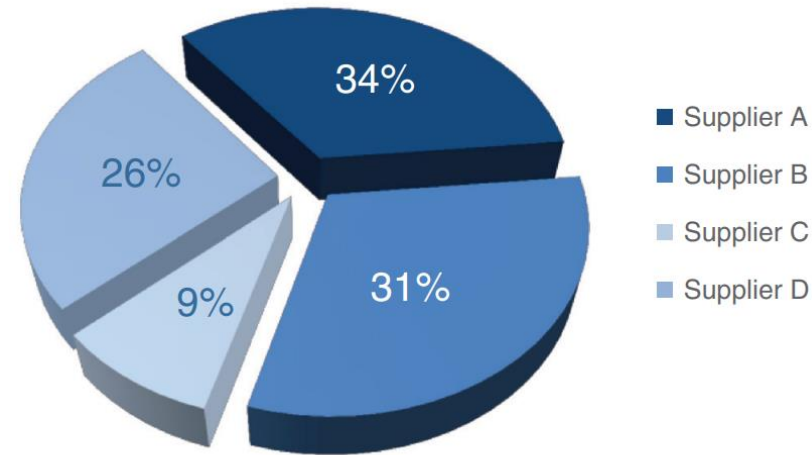


FIGURE 2.22 Pie chart with labeled segments

“Supplier B”—which *looks* largest, at 31%—is actually smaller than “Supplier A” above it, which looks smaller.

Donut charts

- When we use pie charts, we ask our stakeholders/audiences to **compare angles and areas**.
- When we use donut charts, we ask our audience to **compare one arc length to another arc length** – which can be hard to do (a.k.a cognitive load)
- In the image here, how confident do you feel in your eyes' ability to ascribe quantitative value to an arc length (without annotation/labels)?
- If you must use pie charts or donut charts, you need to annotate them/use labels to show values. This might introduce cognitive load /make the charts complex-looking

The donut chart

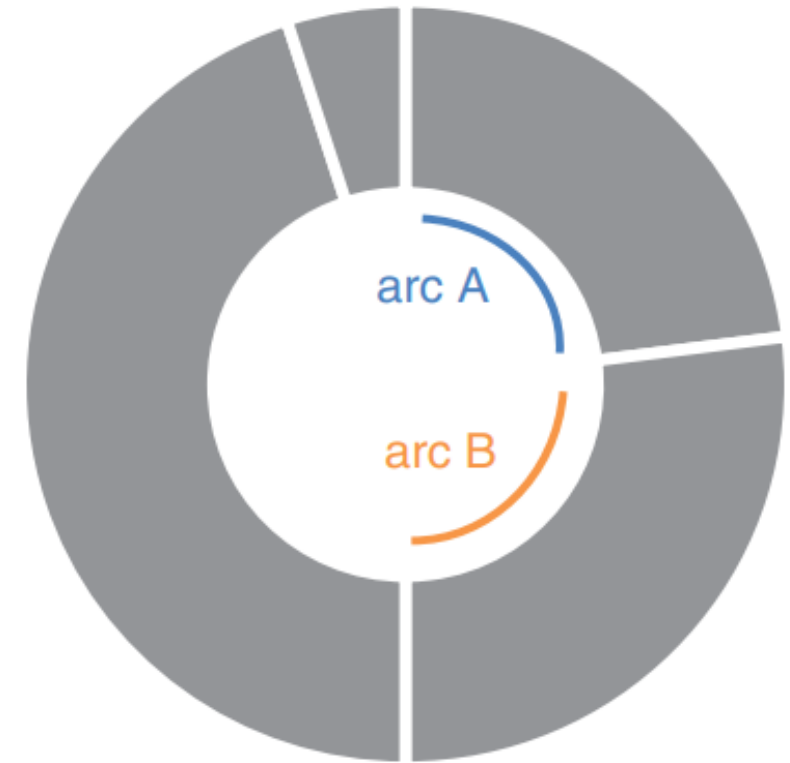


FIGURE 2.24 Donut chart

Alternative to using pie charts/donut charts

- One alternative to using pie/ donut charts is to use a horizontal bar chart
- With bar charts, our eyes compare the end points (easier to do than trying to ascribe quantities to two-dimensional spaces like pie charts)
- Since horizontal bar charts are aligned at a common baseline, **it is easy to assess relative size (a.k.a low cognitive load)**

Supplier Market Share

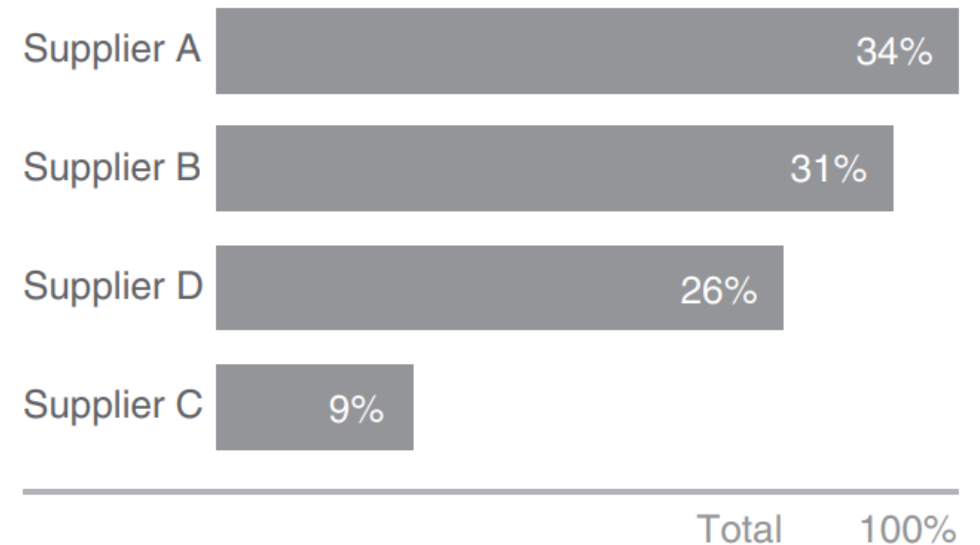


FIGURE 2.23 An alternative to the pie chart

“Never use 3D” – Knafllic (2017)

- The only exception to using 3D visuals is if you are plotting a third dimension or using specialized software (e.g. engineering software) that your stakeholder (e.g. defense contractors, architects) uses to make decisions
- Adding 3D to graphs introduces unnecessary elements like side and floor panels, **which makes it more difficult (i.e. add more cognitive load than improve signal) to read the graph**
- **The example here shows that a simple bar chart is now more complex to read because of the addition of 3D effects**

Number of issues

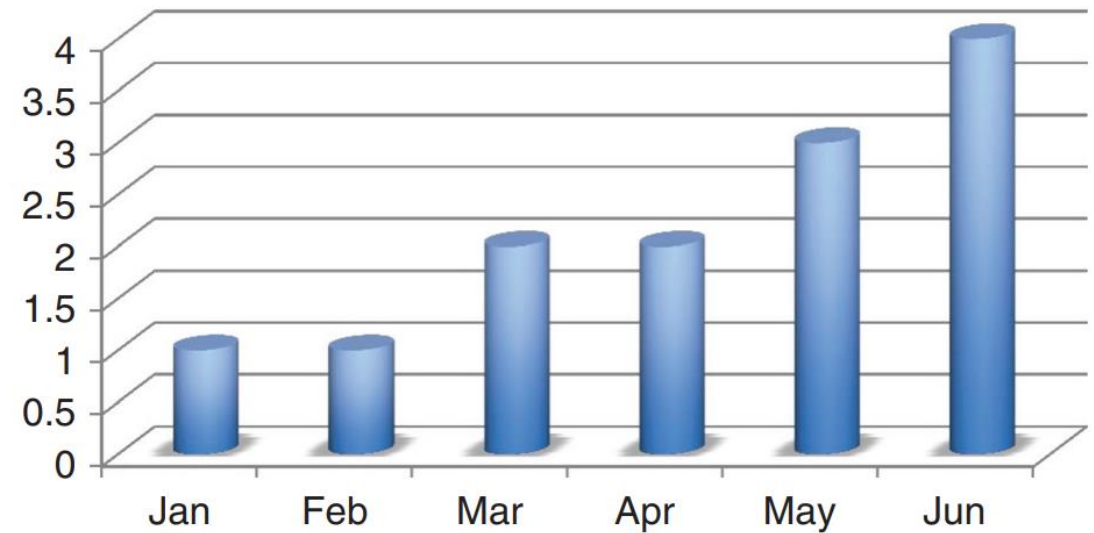


FIGURE 2.25 3D column chart

Should you use a second y-axis on graphs?

- A chart with more than one y-axis can be relatively harder to read
- For attention-deficient stakeholders, interpreting a chart with multiple y-axes takes time – **reading to understand which data should be read against which axis**
- **Multiple y-axis on one chart introduces or increases cognitive load which, in turn, negatively affects effective stakeholder communication**
- **Since most of us read from left to right (Z-manner) you should avoid the use of a secondary or right-hand y-axis**

Secondary y-axis

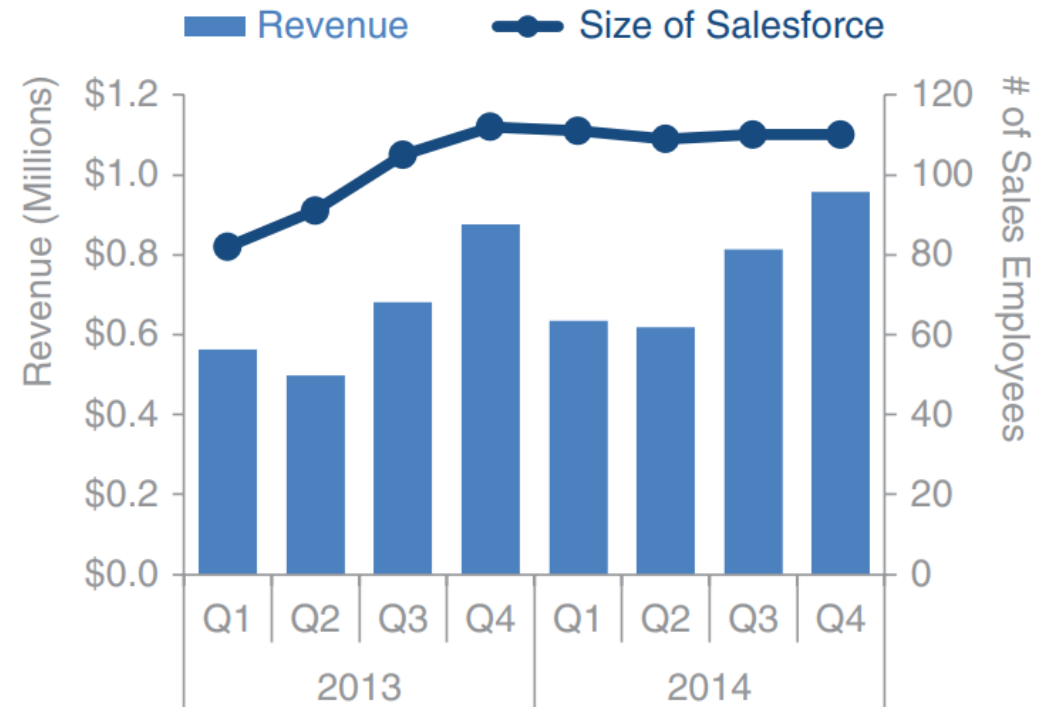


FIGURE 2.26 Secondary y-axis

Alternatives to using a secondary y-axis on one graph

- **Alternative 1:** Do not show the second y-axis. Instead, label the data points that belong on this second y-axis directly
- **Alternative 2:** Pull the graphs apart vertically and have a separate y-axis for each (both along the left) but use the same x-axis across both graphs

Secondary y-axis

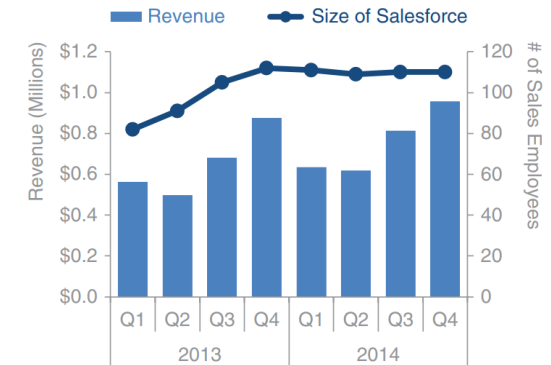


FIGURE 2.26 Secondary y-axis

Alternative 1: label directly



Alternative 2: pull apart vertically



FIGURE 2.27 Strategies for avoiding a secondary y-axis

Is there a “most effective” visual for your situation?

- There is no one “**most effective visual**” for your situation and stakeholders
 - You can create what you think is the best dashboard or visual and still fail to effectively communicate with your attention-deficient stakeholders
- The most effective visual for your stakeholders is the one which:
 - Is easiest for your stakeholders to consume (lowest cognitive load)
 - Enables your stakeholders to retain information the easiest given the context for communication (why?,who?,what?,how?) and your domain expertise
- Remember to **always tailor your data/information visualizations to the needs of your stakeholders** not your own desires
 - Meet with key stakeholders to obtain feedback and modify your charts accordingly **before presenting to a wider audience**

End of presentation